One-to-Many and Many-to-One Relationships

Lecture Notes · Game DB Junior

In this lesson

- There are **four kinds** of relationships between records
- The first two: **one-to-many** and **many-to-one**
- The sign words: **has many** and **has one**

1. Why relationships?

In our first design, records in the different tables are **linked** to each other. Now we start to talk about how to show a relationship between records in our database.

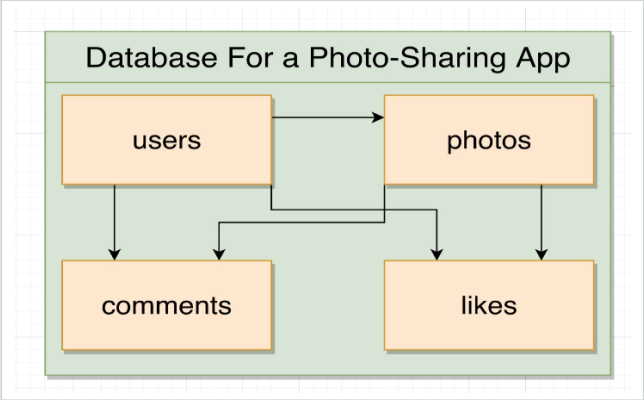


Figure 1 — Our first design

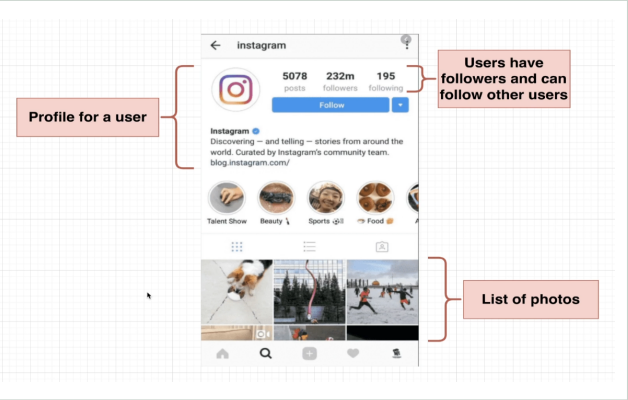


Figure 2 — A user profile page

On the profile page, the user **Instagram** posted some photos. These photos are related to, or owned by, this user. So in our database we need some data that says: **these photos belong to this user**.

To model this, we will learn about **four kinds of relationships**. First we learn the words for them. Then we learn how to build each one in our database, with examples.

2. Kind 1: One-to-Many

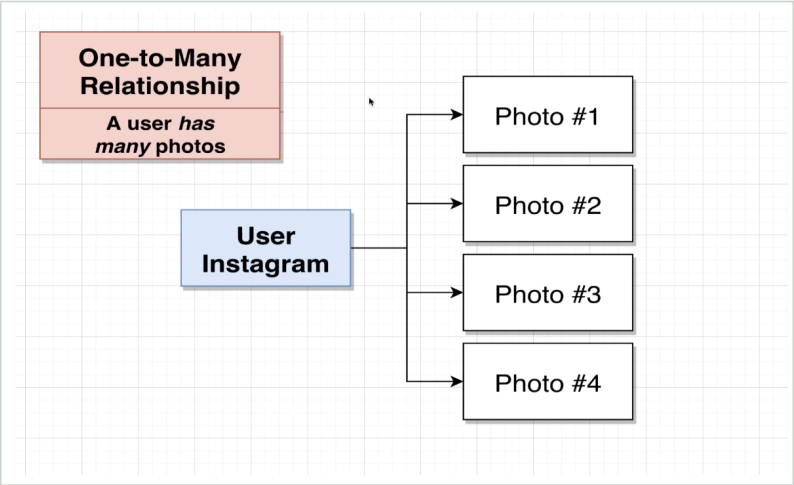


Figure 3 — One-to-many: a user has many photos

- The user Instagram owns **many** photos. The user is tied to these photos.
- This is a **one-to-many** relationship: **one** user has **many** photos.
- In everyday language, when talking to another engineer, you can say: "a user **has many** photos".
- Any time you see the words **has many**, it is a sign of a one-to-many relationship.

3. Kind 2: Many-to-One

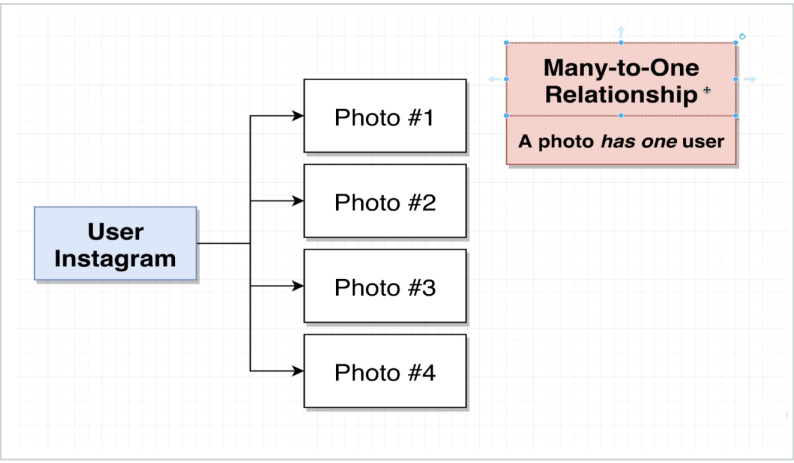


Figure 4 — Many-to-one: a photo has one user

- This is the **same** relationship, looked at from the other side: from the photo.
- Many photos belong to **one** user. Or: a photo **has one** user.
- This is a **many-to-one** relationship.

Key idea: one-to-many and many-to-one are the **exact opposite** of each other. It only depends on which side you look from. From the user: one-to-many. From the photo: many-to-one. You can flip the relationship, and the opposite is still true.

4. Example: photos and comments

On a single post, users can add comments to a photo. How do we describe this relationship?

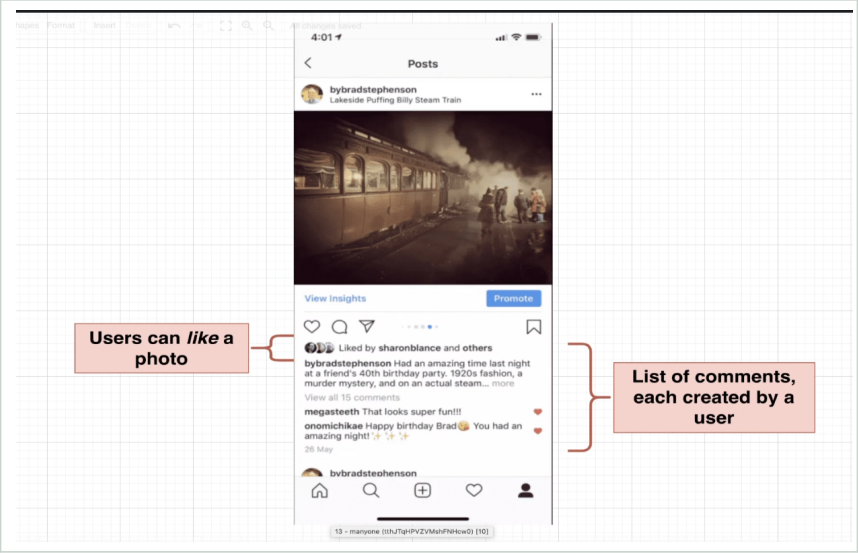


Figure 5 — A single post with comments

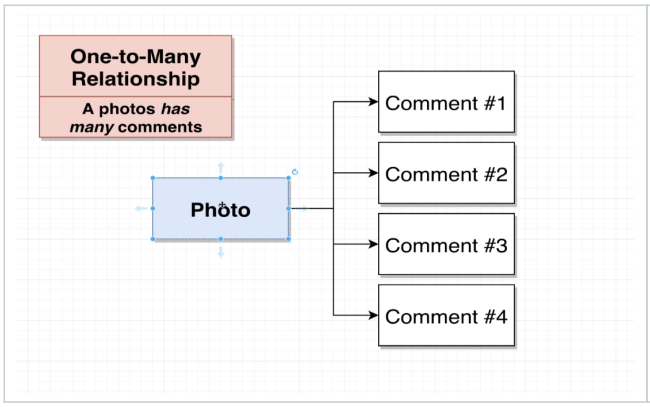


Figure 6 — One-to-many: a photo has many comments

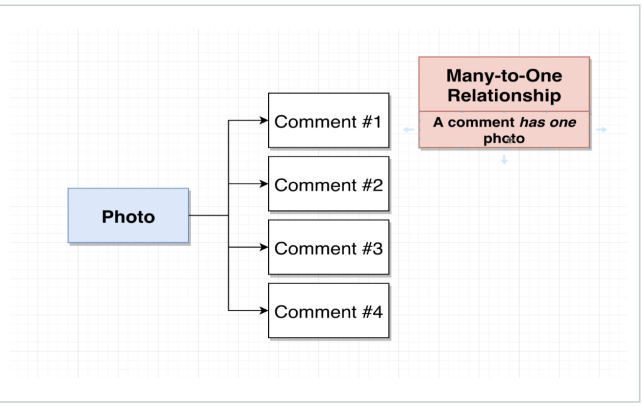


Figure 7 — Many-to-one: a comment has one photo

- From the photo: a photo **has many** comments. This is **one-to-many**.
- From the comment: many comments belong to one photo, or a comment **has one** photo. This is **many-to-one**.

5. More examples

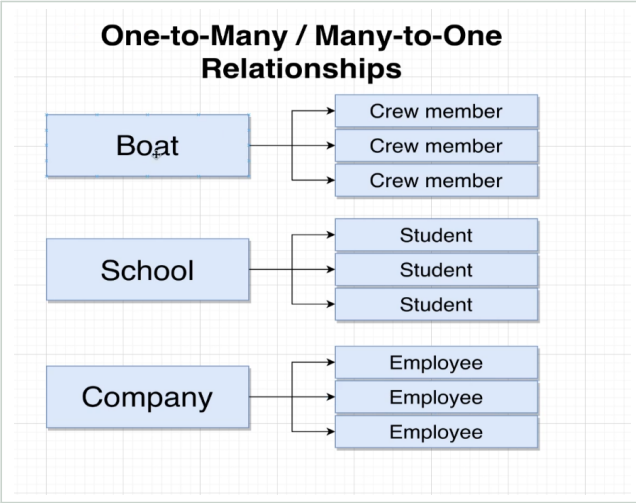


Figure 8 — Boat, school, company

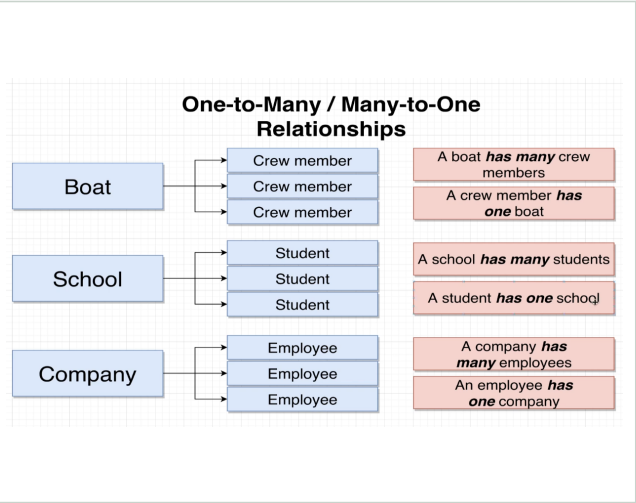


Figure 9 — Has many / has one

One-to-many (left to right)	Many-to-one (right to left)
A boat **has many** crew members	A crew member **has one** boat
A school **has many** students	A student **has one** school
A company **has many** employees	An employee **has one** company

- A crew member is usually on **one boat** at a time.
- Some people go to two schools at the same time, but here we **generalize**: a student is in one school at a time.
- Many people have more than one job, but here we generalize: an employee works for one company at a time.

6. Summary

Idea	Remember
Four kinds	There are four kinds of relationships. This lesson covers the first two
One-to-many	One record has many others. Sign words: **has many**
Many-to-one	Many records belong to one. Sign words: **has one**
Same relationship	One-to-many and many-to-one are opposites. It depends on the side you look from
Examples	user → photos, photo → comments, boat → crew members, school → students, company → employees

Next lesson: the two other kinds of relationships. After that: how to build one-to-many and many-to-one relationships in our database.

Sources
Lecture transcript and lecture images (Figures 1–9)